

Review Problems for Final Exam

1. Evaluate the limit, or explain why the limit does not exist.

(a) $\lim_{x \rightarrow 0^+} (1 + 2 \sin(3x))^{\frac{1}{x}}$

(b) $\lim_{x \rightarrow 0} \frac{\ln(1-x) + x}{x^2}$

(c) $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x^3 + x}$

(d) $\lim_{x \rightarrow \infty} x \ln \left(\frac{x+2}{x} \right)$

2. Find $f'(x)$.

(a) $f(x) = x^3 e^{-x^2}$

(b) $f(x) = \frac{\sqrt{x}}{1+x^2}$

(c) $f(x) = (1+x^8)^x$

3. A box with open top is to have a volume of 8 m^3 . The base is to be square. The material for the base costs twice as much per m^2 as the material for the sides. Find the dimensions of the box that will minimize the total cost of materials. (Assume that there is no waste of materials in constructing the box.)

4. Assume that $f : \mathbb{R} \rightarrow \mathbb{R}$ is twice differentiable and that $f(1) = 5$, $f'(1) = 3$, and $f''(1) = -2$. Define $F : [0, \infty) \rightarrow \mathbb{R}$ by $F(x) = f(\sqrt{x})$. Find $F''(1)$.

5. Find $f^{(31)}(0)$ for the function f defined by $f(x) = x^3 e^{-x^2}$.

6. Find the Taylor series about 0 for the function f defined by

$$f(x) = \ln(\sqrt{1-x^4}), \quad x \in (-1, 1).$$

7. Assume that $\int_0^2 f(x) dx = 3$, $\int_1^2 f(x) dx = 1$, $\int_1^5 f(x) dx = 2$, and $\int_3^5 f(x) dx = 4$.

(a) Find $\int_2^3 f(x) dx$.

(b) Find $\int_6^9 f\left(\frac{x}{3}\right) dx$.

8. Evaluate each of the following indefinite integrals.

(a) $\int x \ln(x^2 + 1) dx$

(b) $\int \frac{x^2}{x+1} dx$

(c) $\int e^{-2x}(e^{7x} + e^{-x}) dx$

(d) $\int \frac{x}{(x^2 + 1)^2} dx$

9. Evaluate each of the following definite integrals.

(a) $\int_0^1 x^3 e^{x^2} dx$

(b) $\int_1^4 \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right)^2 dx$

(c) $\int_0^2 \frac{x^2}{1+x^3} dx$

10. Determine whether or not each improper integral converges. (A quick explanation is sufficient.)

(a) $\int_0^\infty x^{100} e^{-\sqrt{x}} dx$

(b) $\int_0^\infty \frac{x^2}{1+x^3} dx$

(c) $\int_e^\infty \frac{dx}{(\ln x)^{100} \sqrt{x}}$

(d) $\int_e^\infty \frac{dx}{(\sqrt{\ln x}) x^{100}}$

11. Find $F'(x)$ for each of the following. (You may leave un-evaluated integrals in your answer.)

(a) $F(x) = \int_{x^3}^\infty e^{-t^2} dt$

(b) $F(x) = \int_1^{\ln x} \sqrt{t^4 + x^3} dt$

(c) $F(x) = \int_{\sqrt{x}}^{\sin x} \ln(1+t^{10}) dt$

12. Assume that $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ has continuous first-order partial derivatives and that $\vec{\nabla} f(1, 0, 1) = (5, -1, 3)$. Define $\vec{u} : \mathbb{R} \rightarrow \mathbb{R}^3$ by

$$\vec{u}(t) = (e^{2t}, t^3 - 3 \sin t, e^{-t}),$$

and $F : \mathbb{R} \rightarrow \mathbb{R}$ by $F(t) = f(\vec{u}(t))$. Find $F'(0)$.

13. Consider the function f defined by

$$f(t, x, y) = (x^2 + t^2) \ln(1 + x^2 + y^3).$$

Assume that x and y are differentiable functions of t such that when $t = 1$ we have

$$x = 1, \quad y = 2, \quad \frac{dx}{dt} = 3, \quad \frac{dy}{dt} = -1.$$

Find the value at $t = 1$ of the total time derivative $\frac{d}{dt} f(t, x(t), y(t))$.

14. Write an equivalent double integral with the order of integration reversed.

(a) $\int_{-1}^1 \int_{x^2}^1 f(x, y) \, dy \, dx$

(b) $\int_0^6 \int_0^{\frac{1}{3}y} f(x, y) \, dx \, dy$

15. Evaluate each double integral.

(a) $\int_0^1 \int_0^{\sqrt{y}} x^3 y^2 \, dx \, dy$

(b) $\int_0^2 \int_0^{\sqrt{4-x^2}} \sqrt{x^2 + y^2} \, dy \, dx$

16. Solve each of the following initial-value problems.

(a) $y'(t) = e^{2t-y(t)}; \quad y(0) = -1$

(b) $y'(t) + 3y(t) = 5; \quad y(0) = -2$

(c) $y'(t) = 2ty(t) - e^{-t^3}; \quad y(0) = 1$

17. Find the general solution of each of the following.

(a) $t^2 y''(t) - 4t y'(t) + 6y(t) = 0$

(b) $y''(t) - 36y(t) = 0$

(c) $y''(t) - 4y'(t) + 4y(t) = 0$

18. Find $\det(A)$ for each of the following.

(a) $A = \begin{pmatrix} -3 & 2 \\ \sqrt{5} & 4 \end{pmatrix}$

(b) $A = \begin{pmatrix} 2 & 5 & 0 \\ 0 & -2 & 4 \\ 1 & 2 & 1 \end{pmatrix}$

19. Let $A = \begin{pmatrix} -2 & -4 & 2 \\ -2 & 1 & 2 \\ 4 & 2 & 5 \end{pmatrix}$.

Given that

$$\begin{pmatrix} 2 \\ -3 \\ -1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$$

are eigenvectors for A , find the eigenvalues of A as efficiently as you can.

20. Let

$$A = \begin{pmatrix} 1 & 2 & 3 & 6 \\ 0 & -2 & 4 & -3 \\ 1 & 1 & 1 & 4 \\ 4 & -7 & 0 & 5 \\ 3 & -1 & 2 & 2 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 6 & -2 & 4 & 1 & 3 \\ 1 & -1 & 1 & 0 & 4 \\ 3 & -1 & 7 & 1 & 1 \end{pmatrix}$$

Find $\det(AB)$. (There is a very elegant and quick way to do this.)

21. Let $A \in M^{N \times N}$ be given and n be a positive integer. Assume that $A^n = 0$. Show that $I - A$ is invertible and find a formula for $(I - A)^{-1}$.